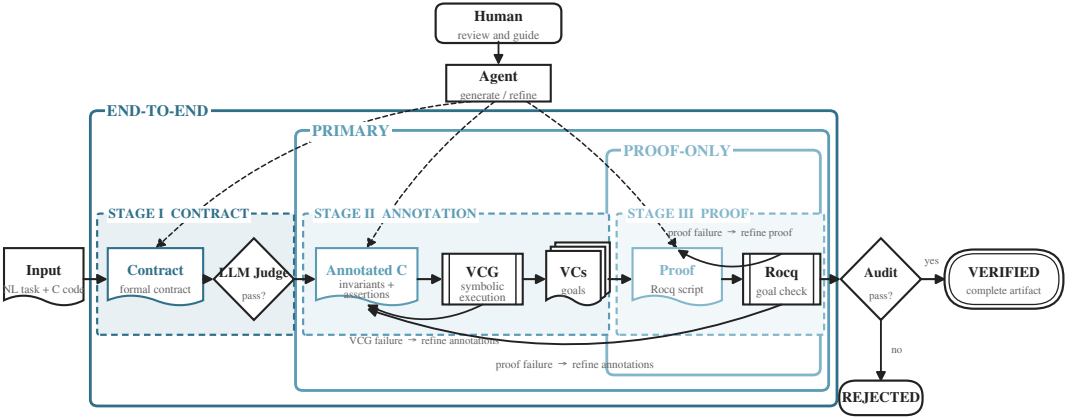
Human
review and guide
Agent
generate / refine
END-TO-END
PRIMARY
PROOF-ONLY
STAGE I CONTRACT
STAGE II ANNOTATION
STAGE III PROOF
Input
NL task + C code
Contract
formal contract
LLM Judge
pass?
Annotated C
invariants + assertions
VCG
symbolic execution
VCs
goals
Proof
Rocq script
Rocq
goal check
proof failure → refine proof
Audit
pass?
yes
VERIFIED
complete artifact
no
REJECTED
VCG failure → refine annotations
proof failure → refine annotations